

1. Data Preprocessing Pipeline

Medical data is often heterogeneous and sparse. The pipeline uses several mathematical steps to ensure data quality.

- **Median Imputation:** Missing values ($X_{i,j}$) are filled using the column median rather than the mean to provide robustness against clinical outliers (e.g., extreme glucose readings). If $X_{i,j}$ is NaN, $X_{i,j} = \text{median}(X_{:,j})$
- **Standard Scaling (Z-score Normalization):** Features are transformed to have a mean (μ) of 0 and a standard deviation (σ) of 1, which is essential for consistent feature comparison in SHAP analysis.

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

- **SMOTE (Synthetic Minority Over-sampling Technique):** To combat class imbalance, SMOTE generates synthetic minority samples (X_{new}) along the line segments joining k-nearest neighbors[cite: 1159, 1161].

$$X_{\text{new}} = X_i + \lambda \cdot (X_k - X_i) \quad \text{where } \lambda \in [0, 1]$$

2. Model Architecture: Random Forest

The engine uses **Random Forest (RF)** classifiers because they perform exceptionally well on the small tabular datasets (300–800 samples) typical in clinical research.

- **Logic:** RF is an ensemble of decision trees where each tree is trained on a bootstrap sample of data[cite: 1177].
- **Gini Impurity:** Splitting at each node is determined by the Gini index (G), which measures the probability of a random sample being incorrectly classified.

$$G(S) = 1 - \sum_k p_k^2 \quad \text{where } p_k \text{ is the proportion of class } k$$

- **Reasoning:** Unlike Neural Networks, RF is scale-invariant, handles missing data natively (after imputation), and is highly resistant to overfitting.

3. Probability Calibration: Platt Scaling

Raw Random Forest probabilities tend to cluster near 0 and 1. **Platt Scaling** is used to "stretch" these outputs into reliable clinical probabilities.

- **Formula:** It wraps the base classifier output ($f(x)$) in a logistic regression model.

$$P_{\text{calibrated}}(y = 1|x) = \frac{1}{1 + e^{-(A \cdot f(x) + B)}}$$

- **The 0.35 Threshold:** A decision threshold of **0.35** is used instead of the default 0.50. This prioritizes **sensitivity** (catching true positive cases), which is critical for medical screening where missing a high-risk patient is a severe failure.

4. Composite Health Score Logic

Individual disease probabilities (h, d, k) are combined into a single 0–100 score.

- **Formula:** Composite Risk = $h \cdot w_{\text{heart}} + d \cdot w_{\text{diabetes}} + k \cdot w_{\text{kidney}}$
- **Health Score:** $(1 - \text{Composite Risk}) \times 100$.
- **Age-Adjusted Ceiling:** For patients ≥ 70 , the maximum score is capped at **95** to reflect the inherent physiological risk of aging.

5. SHAP Explainability & Potential Failures

SHAP (SHapley Additive exPlanations) uses cooperative game theory to assign a contribution value (ϕ_i) to each feature for a specific patient's prediction.

- **Mathematical Foundation:** $\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$
- **Why SHAP may fail:**
 1. **Library Unavailability:** If the `shap` library is not installed in the environment.
 2. **Wrapper Conflict:** SHAP's `TreeExplainer` requires direct access to the base estimator. It may fail if it cannot extract the underlying RF model from the `CalibratedClassifierCV` wrapper.
 3. **API Versions:** Differences in `scikit-learn` versions can break the internal API that retrieves estimators.
 4. **Unsupported Estimators:** While RF is supported, some custom base models may not be compatible with `TreeExplainer` logic.
- **Fallback Logic:** If SHAP fails, the system defaults to **Mean Gini Importance** (Global Feature Importance), which is computationally trivial but lacks the "directionality" (positive/negative impact) of SHAP values.

6. Trend Forecasting: ARIMA

OmniV1 uses **ARIMA(1,1,0)** to predict the next visit's score based on historical momentum.

- **Formula (AR(1) on first differences):** $\Delta y_t = \phi \cdot \Delta y_{t-1} + \epsilon_t$.

- **Reasoning:** Differencing ($d = 1$) makes the series stationary, while the autoregressive component ($p = 1$) captures the persistent clinical "momentum" of a patient's health trajectory